OFFICIAL PRODUCTION RELEASE RUNBOOK

Canine Health & Longevity OS

Professional Step-by-Step Guide & Checklist to Production

EXECUTIVE DEPLOYMENT SPECIFICATION & TARGET SLOs

This production operations guide governs the commercial transition of Canine Health & Longevity OS to a zero-downtime, autoscaling cloud architecture. The system integrates Express.js 4.x, React 19, Google Cloud Run container orchestration, Firebase Firestore security hardening, Microsoft Azure / Edge HD Neural Studio synthesis, and Google Gemini API services. Compliance with the 40 verification criteria is mandatory prior to production traffic cutover.

TARGET SERVICE LEVEL OBJECTIVES

- System Availability: 99.9% Monthly Uptime
- API Latency: P95 < 250ms Across US Regions
- Edge CDN Static Delivery: < 80ms Latency
- TTS Audio Streaming: Initial Byte < 600ms
- Data Durability: Automated Daily Backups (30d)

INFRASTRUCTURE PROFILE

- Runtime: Google Cloud Run (Node 22 Alpine)
- Database: Google Cloud Firestore (Managed)
- Secrets: Google Cloud Secret Manager
- Ingress: HTTPS / Managed TLS 1.3 Certificate
- Autoscaling: 1 to 20 Concurrent Instances

RUNBOOK PHASES & VERIFICATION MODULES

- Phase 1: Environment Variables, Secrets & Configuration Management
- Phase 2: Database Hardening, Firestore Security Rules & Indexes
- Phase 3: Backend Express Hardening & Audio Engine Resilience
- Phase 4: Frontend Optimization, Asset Bundling & Caching Headers
- Phase 5: Multi-Stage Dockerization & Cloud Run Orchestration
- Phase 6: Custom Domain, Managed SSL/TLS & Cloud CDN Edge Setup
- Phase 7: Monitoring, Telemetry, Structured Logging & Alert Policies
- Phase 8: Master Pre-Flight Go / No-Go Checklist (40 Strict Criteria)
- Phase 9: Day-1 Operational Runbook & Rapid Rollback Procedures

Phase 1: Environment Variables, Secrets & Configuration

Production security mandates zero plain-text secrets in source repositories. All server-side credentials must be provisioned via Google Cloud Secret Manager and injected at runtime into Cloud Run containers.

- **NODE_ENV=production:** Enables Express production caching, disables verbose error stack traces.
- **GEMINI_API_KEY:** Injected exclusively into Cloud Run container memory; inaccessible from browser bundles.
- **PORT=3000:** Server listening port matching the internal Cloud Run container routing.
- **CORS & Origin Hardening:** Strictly binds allowed origins to production domain (https://doghealthos.com).

Phase 2: Database Hardening, Firestore Security & Schema Rules

The Firestore database enforces strict User Access Isolation (RBAC) via declarative security rules deployed directly to the production project (ai-studio-dogbreedlifeexpe-6edb2685-ea7c-424d-97f2-ce6b32cfa948).

FIRESTORE SECURITY RULE ENFORCEMENT

• User Isolation: User profiles, dog profiles, and appointment records are strictly constrained by request.auth.uid == userId.
• Admin Boundary: Administrative write privileges for global FAQs and system settings require verified admin claims.
• Automated Backups: Cloud Firestore automated daily backup schedule configured with 30-day retention for point-in-time disaster recovery.

Phase 3: Backend Express Hardening & Audio Engine Resilience

The backend server provides a resilient bridge between client interactions, neural voice generation, and Gemini AI services.

- **Helmet Security Headers:** Enforces CSP, X-Frame-Options, HSTS (31536000s), and nosniff content security headers.
- **Rate-Limiting Protection:** Caps TTS requests at 20 req/min per IP to prevent abusive resource exhaustion.
- **In-Memory Stream Buffer:** Directly buffers EdgeTTS synthesized audio into memory Base64 data without residual disk fragmentation.
- **Health Check Probes:** Provides /healthz and /readyz endpoints returning HTTP 200 OK for Cloud Run ingress probes.

Phase 4: Frontend Optimization, Asset Bundling & Caching Headers

The frontend SPA is compiled into an optimized static distribution via Vite, minimizing bandwidth and maximizing First Contentful Paint (FCP).

- **Code Splitting & Dynamic Chunks:** Heavy visualization libraries (D3, jsPDF) are dynamically chunked to keep entry bundle < 350KB.
- **Immutable Asset Caching:** Hashed bundles (/assets/*.js, *.css) served with max-age=31536000, immutable.
- **Dynamic HTML Freshness:** HTML entry point served with no-cache, no-store, must-revalidate to ensure instant deployment cutovers.

Phase 5: Multi-Stage Dockerization & Cloud Run Orchestration

The application executes inside a hardened, two-stage Docker container based on node:22-alpine.

- **Non-Root Security Context:** Container executes under the non-privileged "node" user, neutralizing container-breakout vectors.
- **Min Instances = 1:** Maintains a warm container instance 24/7 to completely eliminate cold-start latency for end users.
- **Concurrency & Auto-Scaling:** Configured for 80 concurrent requests per container, scaling dynamically from 1 to 20 instances.

Phase 6: Custom Domain, Managed SSL/TLS & Cloud CDN Edge Setup

Global routing leverages Google Cloud Load Balancing with automated multi-region TLS certificates.

- **DNS Records:** Directs apex (@) and app CNAME records to Google Hosted ingress points with 300s TTL.
- **Managed SSL/TLS:** Auto-renewed 2048-bit RSA/ECC certificates with automatic HTTP-to-HTTPS redirect.
- **Cloud CDN Acceleration:** Static UI assets and PDF manuals cached globally across 100+ edge points of presence.

Phase 7: Monitoring, Telemetry, Structured Logging & Alerts

Cloud-native telemetry provides immediate visibility into application health and anomalies. Alerts are configured in Google Cloud Monitoring with email and SMS escalation to anthonytoudou@gmail.com.

- **8.1 Structured JSON Logging:** All server events output standardized JSON payloads with severity levels for Google Cloud Logging.
- **8.2.1 High 5xx Error Rate Alert:** Threshold: > 1.0% of requests returning 5xx status codes over 5m rolling window (Action: Pager / Immediate
- **8.2.2 Elevated Latency Alert:** Threshold: 95th percentile request latency exceeding 1,200ms over 10m (Action: High-priority notification).
- **8.2.3 Memory Exhaustion Alert:** Threshold: Memory utilization exceeding 85% of allocated 1GB (Action: Auto-scale trigger & memory review)
- **8.2.4 Firebase Security Rule Rejections:** Threshold: > 10 permission-denied occurrences within 5 minutes (Action: Security audit alert).

Phase 8: Master Pre-Flight Go / No-Go Checklist (40 Criteria)

Every single criterion below must be verified and checked off prior to opening production traffic:

- ☐ **[1]** Zero hardcoded API keys or secrets in Git repository.
- ☐ **[2]** GEMINI_API_KEY stored exclusively in Google Cloud Secret Manager.
- ☐ **[3]** firestore.rules deployed and verified (user isolation strictly enforced).
- ☐ **[4]** Helmet HTTP security headers active (X-Frame-Options, CSP, HSTS).
- ☐ **[5]** API rate limiting active on /api/gemini/tts and public routes.
- ☐ **[6]** CORS policy restricted to production domain origins.
- ☐ **[7]** Client distribution bundle verified clean of private env vars.
- ☐ **[8]** Express request payload size limits enforced (1mb max).
- ☐ **[9]** Breed selector autocompletes accurately across all 200+ AKC breeds.
- ☐ **[10]** Chronological age decimal calculation accurate for puppies and seniors.
- ☐ **[11]** Imperial (lbs) and Metric (kg) toggle converts without distortion.
- ☐ **[12]** D3 Growth Chart renders sigmoidal curves, P10-P90 bands, and milestones.
- ☐ **[13]** D3 Longevity Impact curve accurately reflects Purina Lifespan model.
- ☐ **[14]** Weight Sensitivity Slider recalculates lifespan in real time.

- [ ] **[15]** Body Condition Scoring (BCS 1-9) renders silhouettes and actions.
- [ ] **[16]** Historical Weight Logger successfully records, edits, and deletes logs.
- [ ] **[17]** Vet Appointments scheduler creates, edits, and completes visits.
- [ ] **[18]** Calendar .ics export generates valid events for Apple & Google Calendar.
- [ ] **[19]** All 8 veterinary specialist voice personas audition cleanly.
- [ ] **[20]** Pitch resonance controls (-2 Hz, 0 Hz, +2 Hz) modulate frequency.
- [ ] **[21]** Live glowing transcript highlights synchronize with spoken audio.
- [ ] **[22]** Animated waveform visualizer functions smoothly without stutter.
- [ ] **[23]** Audio Master MP3 export generates downloadable, playable audio.
- [ ] **[24]** Graceful acoustic chime fallback executes if network drops occur.
- [ ] **[25]** Export Full Clinical PDF compiles clean multi-page consultation doc.
- [ ] **[26]** UI responsive layouts verified across Mobile (375px) and Desktop (1440px).
- [ ] **[27]** Dark and light mode contrast verified (WCAG AA compliant).
- [ ] **[28]** Printable clinical care sheets format cleanly without ink waste.
- [ ] **[29]** Official User Manual PDF serves and downloads cleanly via web route.
- [ ] **[30]** Pet profile modal saves custom names, microchips, and dates reliably.
- [ ] **[31]** npm run lint (tsc --noEmit) passes with 0 errors.
- [ ] **[32]** npm run build succeeds cleanly with optimal code-splitting.
- [ ] **[33]** Cloud Run minimum instances set to 1 (eliminating cold starts).
- [ ] **[34]** Cloud Run memory configured to 1Gi and CPU to 1.
- [ ] **[35]** Liveness probe /healthz responds with HTTP 200 OK.
- [ ] **[36]** Daily automated Firestore backups scheduled with 30-day retention.
- [ ] **[37]** SSL/TLS certificates active with A+ rating on Qualys SSL Labs.
- [ ] **[38]** Cloud Monitoring alerts active for 5xx errors and latency spikes.
- [ ] **[39]** DNS propagation verified globally across major nameservers.
- [ ] **[40]** Executive Sign-off: Final release approval granted by Lead Engineer.

Phase 9: Day-1 Operational Runbook & Rapid Rollback Procedures

Standard operating procedures for immediate post-launch operations and critical incident response.

EMERGENCY ONE-COMMAND ROLLBACK PROTOCOL (< 15 SECONDS)

If a Sev-1 defect occurs in production, immediately shift 100% of live traffic back to the prior stable revision:

gcloud run services update-traffic doghealthos-prod --to-revisions=doghealthos-prod-00012-xyz=100

This instantly redirects user requests to the previous immutable container without rebuilding or restarting the cluster.